sasfit-inversion Documentation

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Standalone Python library (+ a Qt GUI) for solving Fredholm integrals of the first kind arising in small-angle scattering analysis: recovering a **pair-distance distribution function (PDDF)** p(r) or a **size distribution** from measured I(q).

This is the standalone PDF-report build (narrative pages only – see `../index.rst` for the full docs build, which also includes the API reference).

CHAPTER

ONE

INSTALLATION

From the project root:

```
pip install -e ".[gui,dev]"
```

Install just the library, without the Qt/GUI dependencies (`PySide6`, `matplotlib`), with:

```
pip install -e .
```

1.1 Requirements

- Python >= 3.10
- `numpy`, `scipy` (core library)
- `PySide6`, `matplotlib` (GUI only, the `[gui]` extra)
- `pytest` (the `[dev]` extra, for running `tests/`)
- `sphinx`, `furo`, `sphinx-autodoc-typehints` (the `[docs]` extra, for building this documentation)

1.2 Building this documentation

```
pip install -e ".[docs]"
cd docs
make html         # Linux/macOS
make.bat html     # Windows
```

Open `docs/_build/html/index.html` afterwards.

CHAPTER

TWO

USAGE (LIBRARY)

2.1 Minimal example

```
import numpy as np
from sasfit_inversion.io_utils import load_sas_data
from sasfit_inversion.background import fit_background
from sasfit_inversion.kernel_registry import KERNEL_REGISTRY
from sasfit_inversion.kernels import build_size_distribution_kernel
from sasfit_inversion.solver_registry import run_solver

data = load_sas_data("tests/data/test.dat")

# flat background from a fit of backgr + c*q^-alpha (only backgr is
# subtracted, see the theory chapter)
bg = fit_background(data.q, data.I, q_min=0.094, q_max=0.385, dI=data.dI)
b = data.I - bg.evaluate(data.q)

# grid: Dmax = pi/q_min, N = Shannon number
dmax = np.pi / data.q.min()
n = int(round((data.q.max() - data.q.min()) * dmax / np.pi))
r = np.linspace(0.0, dmax, n)

f = KERNEL_REGISTRY["sinc_4pi"].func            # 4*pi*j0(qr), PDDF
A = build_size_distribution_kernel(data.q, r, f, alpha=0.0)

res = run_solver("bayesian_evidence", A, b, data.dI)
print(res.chi2_r_history[-1], res.diagnostics["lambda_selection"])
p_r, p_sigma = res.x, res.diagnostics["x_sigma"]
```

`run_solver(key, A, b, dI, spline=None, **options)` is the single entry point for all solvers. It returns a `SolverResult` with `x` (the distribution on the grid), `fitted_b` ($\mathbf{Ax}$), `chi2_r_history`, `converged`, `n_iterations` and `diagnostics` (`lambda_selection` text, and depending on the solver `x_sigma`, `x_cov`, `lambda`, `log_evidence_normalized`, ...).

2.2 Solvers and options

key	options (first = default)	notes
`em_discrepancy`	`selection`: `"discrepancy"` | `"lcurve"`	EM + smoothing; non-negative kernel
`maxent_em`	`prior`: `"adaptive"` | `"constant"`	SASfit EM + MaxEnt; non-negative kernel
`hansen_maxent`	`target_chi2_r` (1.0)	Hansen functional, $x > 0$
`bayesian_evidence`	`penalty`: `"hansen_eq19"` | `"hansen_printed"` | `"second_derivative"`; `noise_scale`: `"given"` | `"estimate"`; `target_chi2_r` (None)	recommended for the PDDF; Laplace error band, $\ln Z$
`general_tikhonov_gcv`		2nd difference, GCV
`tikhonov_standard_gc`		$\mathbf{L} = \mathbf{I}$, GCV
`tsvd_gcv`		truncated SVD, GCV
`arlsnn`		ARLS with $x \geq 0$, no parameter
`em_general_signed_ke`	`regularization`: `"smoothing"` | `"maxent_adaptive"` | `"maxent_constant"`	EM for signed kernels (PDDF)

Example: `run_solver("bayesian_evidence", A, b, dI, penalty="hansen_printed", target_chi2_r=1.0)`.

Old keys still work: `SOLVER_ALIASES` maps e.g. `em_lcurve` to `("em_discrepancy", {"selection": "lcurve"})` and `bayesian_evidence_hansen` to `("bayesian_evidence", {"penalty": "hansen_printed"})`; the retired solvers (`bayesian_evidence_spline`, `em_discrepancy_spline`, `bayesian_evidence_two_segment`, `bayesian_evidence_hann_tapered`) are kept in `RETIRED_SOLVERS`. `get_solver(key)` resolves any key.

2.3 Kernel and solver compatibility

- `sinc_4pi` (signed): `bayesian_evidence`, the Tikhonov/TSVD solvers, `arlsnn` (only if $p \geq 0$ is acceptable) and `em_general_signed_kernel`. `em_discrepancy` and `maxent_em` diverge with a signed kernel.
- `sphere_rg` (non-negative): every solver; `em_discrepancy` is the best validated. Use $\alpha = 6$.

2.4 B-spline representation

Any solver can solve for B-spline coefficients instead of point values:

```
from sasfit_inversion.kernels import quadrature_points
from sasfit_inversion.solver_registry import SplineOptions

```

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```
nq = quadrature_points(dmax, data.q.max(), n)          # fine grid
A_quad = build_size_distribution_kernel(data.q, np.linspace(0, dmax, nq), f)
opt = SplineOptions(n_coeffs=20, spacing="log", A_quad=A_quad)
res = run_solver("em_general_signed_kernel", A, b, data.dI, spline=opt,
                 regularization="maxent_adaptive")
# res.x is p(r) on the grid r; coefficients in res.diagnostics["spline_
↪coefficients"]
```

`A` (the kernel on the display grid `r`) only defines the display grid; the solver works on `A_quad @ B`.

2.5 Non-uniform grids and interpolation

```
from sasfit_inversion.kernels import make_r_grid, build_kernel_on_nodes

r = make_r_grid(dmax, 30, "log")                       # uniform | quadratic |␣
↪log
A = build_kernel_on_nodes(data.q, r, f, interpolation="cubic")   # or "linear"
res = run_solver("bayesian_evidence", A, b, data.dI)
```

The unknowns are the values at `r`; between the nodes $p(r)$ is linear or a not-a-knot cubic spline and the kernel is integrated on a fine grid. Use `"linear"` with solvers that need $x \geq 0$.

2.6 Uncertainty

```
from sasfit_inversion.uncertainty import bootstrap_uncertainty

solve = lambda b_rep: run_solver("bayesian_evidence", A, b_rep, data.dI)
boot = bootstrap_uncertainty(solve, b, data.dI, n_boot=50, center=res.fitted_b)
boot.x_mean, boot.x_std, boot.n_failed
```

Always pass `center=res.fitted_b` (see *Uncertainty of the recovered distribution*).

2.7 Data without error bars

```
from sasfit_inversion.goodness_of_fit import estimate_errors, fit_statistics

d = load_sas_data("tests/data/test_2col.dat")          # q, I only: d.dI is␣
↪None
sigma = estimate_errors(d.q, d.I, "poly")             # 5-point cubic fits;
                                                       # or "scatter", "poisson
↪", "relative"
res = run_solver("bayesian_evidence", A, d.I - 0.1, sigma,
                 noise_scale="estimate")               # overall level by the␣
↪evidence
res.diagnostics["noise_scale"]                         # s, errors used =␣
↪s*sigma
```

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```
fit_statistics(d.I, res.fitted_b + 0.1, sigma)          # chi2_r, Pearson,␣
↪deviance, KL
```

2.8 Size distributions

```
from sasfit_inversion.kernels import recover_N_from_x

R = np.linspace(0.0, dmax / 2, 150)
A = build_size_distribution_kernel(data.q, R, KERNEL_REGISTRY["sphere_rg"].func,
↪ alpha=6.0)
res = run_solver("em_discrepancy", A, b, data.dI)
N = recover_N_from_x(res.x, R, alpha=6.0)       # number distribution
V = res.x * R**-3.0                              # volume-weighted (x = N R^6)
```

2.9 Tests

The tests are plain scripts (run from the project directory):

```
python tests/test_em_general_basic.py
for f in tests/test_*.py; do python "$f"; done
QT_QPA_PLATFORM=offscreen python tests/test_gui_smoke.py
```

CHAPTER

THREE

USING THE GUI

3.1 Start

```
python src/gui/app.py
# or, after an editable install:
sasfit-inversion-gui
```

3.2 Layout

The window has a control column on the left and two plots on the right: $I(q)$ with error bars, background and fit (log–log) at the top, the recovered distribution with its uncertainty band and a dotted zero line at the bottom.

The control column contains three groups (*Data*, *Background*, *Inversion*) in a scroll area and, below it, the **info panel**. The border between them is a splitter: drag it to change the height of the info panel, like the border between the two plots.

Info panel

All explanations and all detailed results are shown in this single panel instead of in tooltips or long text blocks. It always shows the text of the control under the mouse pointer (the control or its label); with an open drop-down list it explains the highlighted entry (e.g. each solver or kernel). Moving away shows a short default text. Until the splitter is dragged, the panel is kept just large enough for the longest text at the current width, so the layout does not jump while the mouse moves; it never grows beyond that height, and if it is made smaller its text scrolls. Dragging it back to full height re-enables the automatic height. The static texts live in `src/gui/help_texts.py` (Qt rich text, one entry per control, solver, kernel and solver option).

The control column itself only shows one-line summaries; the details appear in the info panel right after the action and again whenever the mouse is over the summary line:

- **Data** (after loading): file and folder, number of points and skipped lines, q range and spacing (points per decade for log-spaced data), I, ΔI and $\Delta I/I$ ranges, $D_{max} = \pi/q_{min}$, the resolution π/q_{max}, the Shannon number and the number of data points per Shannon channel overall and in the upper third of the q range (with a hint to use B-splines if it is below one).
- **Background** (after a fit or with a fixed value): model, fit range and number of points, all parameters with errors, χ^2_r of the fit, the subtracted constant and its fraction of $I(q_{max})$, and notes on extrapolation and degenerate parameters.

- **Result** (after a solve): χ^2_r and its mean per third of the q range, Pearson χ^2/M, Poisson deviance (G-test)/M and the KL divergence of the normalized curves, the estimated error scale s (Bayesian evidence, option *Error bars: overall scale estimated*), solver and options, how the regularization strength was chosen (λ, N_g, $\ln Z$, ...), the representation (grid, interpolation, B-splines, quadrature points), kernel and α, iterations, computing time, derived quantities (PDDF: $I(0) = 4\pi \int p\,dr$, $R_g^2 = \int p\,r^2 dr / (2\int p\,dr)$ and the position of the maximum; size distributions: maximum of $x(r)$, number-weighted mean radius and maximum of the volume distribution), the type of error band, bootstrap statistics and warnings.

The current suggestions for r_{max} and the number of grid points are appended to the corresponding control texts.

3.3 Workflow

1. **Load data file** – three columns q, I, ΔI. r_{max} and the number of grid points are suggested ($D_{max} = \pi/q_{min}$, Shannon number), and are re-suggested when the kernel changes. **Error bars** – from the file's ΔI column, or (always for files without one) an error model: estimated from local cubic fits through 5 points (default) or from neighbour interpolation, combined over a window of points, Poisson ($\sigma^2 = I$, counts only) or a relative error in percent (see *Forward model, data and information content*). The data info shows the ratio of the cubic-fit estimate to the file errors as a check.
2. **Background** – fit $I_{bg} + c\,q^{-\alpha}$ in a q range (only I_{bg} is subtracted) or enter a fixed value.
3. **Kernel** – $4\pi j_0(qr)$ for the PDDF (no α), or the sphere kernel for a radius distribution ($\alpha = 6$ default).
4. **Grid** – r_{max}, number of points, spacing (uniform, quadratic, logarithmic) and interpolation between the points (linear or cubic). Spacing and interpolation are hidden in B-spline mode; interpolation is fixed to linear for the solvers that need $x \geq 0$.
5. **Solver** and its options (appear only for the solver that has them), optionally "Force $\chi^2_r = 1$" (Bayesian evidence).
6. **B-splines** – "Represent p(r) by B-splines (any solver)", number of coefficients and knot spacing.
7. **Monte Carlo uncertainty** – parametric bootstrap with N replicates.
8. **Solve**.

3.4 Result line

The one-line summary below *Solve* shows χ^2_r, the number of bootstrap replicates if used, and a warning sign if something needs attention; everything else is in the info panel (see above). Warnings include a solver that needs a non-negative kernel used with a signed kernel, and a $\chi^2_r = 1$ target that was not reachable.

3.5 Plots

The distribution plot shows the MAP solution with the analytic $\pm 1\sigma$ band for the Bayesian-evidence solver, or the bootstrap mean with its $\pm 1\sigma$ band and the single solve (dashed) when

the Monte Carlo option is used. The y axis is $p(r)$ for the PDDF kernel and $x(r) = N(r)\, r^{\alpha}$ for size distributions. The zero line style is set by `ZERO_LINE_STYLE` in `app.py`.

CHAPTER

FOUR

THEORY

This part of the documentation describes the mathematics behind every step of an inversion with *sasfit-inversion*: the forward model, how the integral is turned into a linear system, how that system is regularized, how the regularization strength is chosen, and how the uncertainty of the result is estimated. The last chapter records the investigations that led to the present set of methods (what was tried, what failed and why).

4.1 Forward model, data and information content

4.1.1 The Fredholm equation

All inversions in this package solve a Fredholm integral equation of the first kind

$$I(q) - I_{bg} \;=\; \int_0^{r_{max}} K(q,r)\, x(r)\, \mathrm{d}r, \tag{4.1}$$

for the unknown function $x(r)$, given M noisy samples $b_i = I(q_i) - I_{bg}$ with uncertainties σ_i. The kernel K is smooth in r, so the integral operator damps high-frequency components of x strongly; inverting it amplifies noise without bound. The problem is *ill-posed* and every method below is a way of adding prior information (smoothness, positivity, entropy, a reduced basis) to obtain a stable, physically meaningful solution.

4.1.2 Kernels

Two kernels are registered in `sasfit_inversion.kernel_registry.KERNEL_REGISTRY`.

Pair-distance distribution (`sinc_4pi`)

For a particle with excess scattering length density $\Delta\rho(\mathbf{r})$ the orientationally averaged intensity is the Debye/Fourier-sine relation

$$I(q) = 4\pi \int_0^{D_{max}} p(r)\, j_0(qr)\, \mathrm{d}r, \qquad j_0(x) = \frac{\sin x}{x}, \tag{4.2}$$

with $p(r) = r^2\,\tilde{\gamma}(r)$, $\tilde{\gamma}$ the autocorrelation of $\Delta\rho$. Hence $K(q,r) = 4\pi j_0(qr)$; the factor r^2 is part of the unknown $p(r)$. Properties used throughout:

- $p(0) = 0$ (it vanishes like r^2) and $p(r) = 0$ for $r \geq D_{max}$; the r grid therefore always starts at $r = 0$.
- The kernel is **signed**: $j_0(qr) < 0$ for $\pi < qr < 2\pi$, etc. Methods that require a non-negative kernel (plain EM, EM + MaxEnt) are not applicable.
- $p(r)$ may itself be negative for particles with an internal contrast that changes sign (core–shell, contrast variation). No positivity is imposed by default.

- There is no size weighting (`uses_alpha=False`).

Sphere size distribution (`sphere_rg`)

For a dilute polydisperse system of homogeneous spheres of radius R with number distribution $N(R)$

$$I(q) = \int_0^\infty N(R)\, R^6 \left[\frac{3\,(\sin qR - qR\cos qR)}{(qR)^3}\right]^2 \mathrm{d}R, \tag{4.3}$$

up to the constant $(4\pi/3)^2\Delta\rho^2$. The factor $R^6 \propto V^2$ must be kept: it sets the *relative* weight of small and large particles. The kernel is non-negative. The variable r of the grid is the **radius**, so the suggested r_{max} is $D_{max}/2$.

Size weighting

Following Kohlbrecher & Bressler (2022, eq. 32–34) the unknown is rescaled as

$$x(R) = N(R)\, R^\alpha, \qquad K_\alpha(q,R) = R^{-\alpha} K(q,R), \tag{4.4}$$

which leaves (4.1) unchanged. $\alpha = 0, 3, 6$ give number, volume and volume-squared (intensity) weighted distributions. For the sphere kernel $\alpha = 6$ divides the R^6 out of the kernel; with $\alpha = 0$ the columns of $\mathbf{A}$ span about 15 decades and the EM iteration fails. At $R = 0$ the column is set to zero for $\alpha > 0$. `sasfit_inversion.kernels.recover_N_from_x()` undoes the substitution.

4.1.3 Background

`sasfit_inversion.background.fit_background()` fits

$$I(q) \approx I_{bg} + c\, q^{-\alpha_{bg}}$$

by weighted non-linear least squares in a chosen q range. **Only the constant** I_{bg} is subtracted (`BackgroundFitResult.evaluate` deliberately evaluates the power law with $c = 0$); the power law only serves to separate the flat background from the decaying particle scattering in the fit. `extrapolation_risk` measures in units of the fit range's own log width how far the data extend outside the fit range, because a power law extrapolated by a decade can exceed the signal. A fixed, user-given constant can be used instead.

4.1.4 Information content and the choice of the grid

Maximum dimension

Glatter's consistency condition $q_{min} \le \pi/D_{max}$ gives the largest dimension the low-q end of the data can constrain. The GUI suggests

$$D_{max} \approx \frac{\pi}{q_{min}}, \qquad r_{max} = \begin{cases} D_{max} & \text{PDDF} \\ D_{max}/2 & \text{sphere radius} \end{cases}$$

For `tests/data/test.dat` ($q_{min} = 0.004016\ \mathrm{nm}^{-1}$) this is $D_{max} \approx 782$ nm. Larger values let the solver "improve" χ^2 by fitting noise at mid q.

Shannon sampling

$p(r)$ on $[0, D_{max}]$ and $I(q)$ are a Fourier pair. The sampling theorem says that $I(q)$ is fully determined by its values at the Shannon points $q_k = k\pi/D_{max}$, and $p(r)$ measured up to q_{max} is resolved on a length scale

$$\Delta r_{Shannon} = \frac{\pi}{q_{max}}.$$

The number of independent parameters (Shannon channels) in the data is

$$N_s = \frac{(q_{max} - q_{min})\, D_{max}}{\pi}, \tag{4.5}$$

which the GUI suggests as the number of r grid points (test.dat: $\pi/q_{max} = 8.2$ nm, $N_s \approx 95$ for the PDDF, 47 for the sphere radius).

Log-spaced data

Most SAS data are equidistant in $\log q$. The number of data points per Shannon channel is then

$$\frac{\mathrm{d}n}{\mathrm{d}k} = \frac{\pi}{D_{max}} \frac{\mathrm{d}n}{\mathrm{d}q} = \frac{\pi}{D_{max}} \frac{1}{q\,\Delta \ln q},$$

i.e. it decreases like $1/q$. On test.dat it drops to about 0.45 points per channel at high q. A model with one free parameter per channel (a point grid with $N \approx N_s$) can then follow every single high-q point, including its noise – this was the "high-q overfitting" analysed in *Development history and findings*, and it is the main reason for the reduced B-spline basis (*Discretization of the integral*).

4.1.5 Goodness of fit

$$\chi^2_r = \frac{1}{M} \sum_{i=1}^{M} \frac{[(\mathbf{Ax})_i - b_i]^2}{\sigma_i^2}$$

is used as the fit criterion (target value 1 in the discrepancy principle). For diagnosing *where* a fit is too good or too bad, the residuals are split into the lower, middle and upper third of the q points and the mean of $[(\mathbf{Ax})_i - b_i]^2/\sigma_i^2$ is reported per third ("low/mid/high-q χ^{2}"); a value far below 1 in one region means that the model follows the noise there. Data without error bars

———————————

Every solver needs σ_i (weights, χ^2_r, discrepancy principle). For files without a ΔI column – or to check the given one – `sasfit_inversion.goodness_of_fit.estimate_errors()` provides three error models.

Local cubic fits (default without ΔI)

For every point a cubic in $\ln I$ versus $\ln q$ is fitted by least squares to the point and its two neighbours on each side (`local_polynomial_scatter()`). With $n = 5$ points and $p = 4$ parameters the residual sum of squares has $\nu = n - p = 1$ degree of freedom, and

$$s_i^2 = \frac{\mathrm{RSS}_i}{\nu} = \frac{1}{n-p} \sum_{k=i-2}^{i+2} [\ln I_k - P_i(\ln q_k)]^2$$

is an unbiased estimate of the relative variance ε_i^2. (The plain mean square deviation RSS/n is too small by $\nu/n = 1/5$.) The cubic removes the local shape of the curve up to third order. For equidistant points the single residual degree of freedom is proportional to the fourth difference, $s_i \propto |y_{i-2} - 4y_{i-1} + 6y_i - 4y_{i+1} + y_{i+2}|/\sqrt{70}$.

A single s_i is $|z|$ of one Gaussian variable ($s_i^2 \sim \varepsilon^2 \chi^2_1$): unbiased in the variance, but its 5–95 % range is 0.06–2 ε, and weights $1/\sigma_i^2$ built from it can be wrong by orders of magnitude. The estimates are therefore combined over a window of N points (default 15) by a running median, scaled to be unbiased for Gaussian noise,

$$\hat{\varepsilon}_i = \frac{\mathrm{median}_{j \in \mathcal{W}_i}\, s_j}{\sqrt{\mathrm{median}(\chi^2_\nu)/\nu}} \quad (= 1.4826\,\mathrm{median}\, s_j \text{ for } \nu = 1), \qquad \sigma_i = \hat{\varepsilon}_i\, |I_i|.$$

$N = 1$ is the plain per-point estimate. Near the ends the first or last five points are used.

Neighbour interpolation

Each point is compared with the line through its two neighbours in $\ln I$ versus $\ln q$,

$$d_i = \ln I_i - [w_i \ln I_{i-1} + (1 - w_i) \ln I_{i+1}], \qquad w_i = \frac{\ln q_{i+1} - \ln q_i}{\ln q_{i+1} - \ln q_{i-1}},$$

$\mathrm{Var}(d_i) = \varepsilon^2(1 + w_i^2 + (1 - w_i)^2)$, combined the same way: $\hat{\varepsilon}_i = 1.4826\,\mathrm{median}_{\mathcal{W}_i}|d_j|/\sqrt{1 + w_j^2 + (1 - w_j)^2}$. Only linear trends are removed, so curvature (steep slopes, form-factor minima) inflates the estimate.

Comparison (median of $\hat{\sigma}/\sigma_{true}$, window 15; synthetic monodisperse spheres with known noise on 91 or 200 log-spaced points, 40 noise realizations; test.dat against its file errors):

data	5-point cubic	3-point neighbour
spheres R = 15, 5 %, 91 points	1.06	1.10
spheres R = 30, 3 %, 200 points	1.09	1.16
spheres R = 30, 3 %, 91 points	1.22	1.28
spheres R = 60, 1 %, 91 points	1.60	1.91
test.dat vs file ΔI	1.20	1.18

The remaining overestimate comes from the sharp minima of monodisperse spheres (few points, low noise); with a smooth curve and 3 % noise the cubic gives $\langle s^2\rangle/\sigma^2 = 1.015$ per point and a median ratio of 1.03 combined.

Poisson

$\sigma_i^2 = I_i$, valid only for intensities in counts.

Relative error

$\sigma_i = r\,|I_i|$ with a given r.

If the *shape* of σ_i is right but its overall level is not (estimated or guessed errors), the Bayesian-evidence solver can estimate an overall factor (*Regularization and choice of the regularization strength*, "Unknown error scale").

4.1.6 Alternative fit statistics

`fit_statistics()` reports, for the measured and fitted **total** intensities I_i, F_i (background included, so that both are positive):

$$\chi^2_r = \frac{1}{M}\sum_i \frac{(I_i - F_i)^2}{\sigma_i^2}, \qquad \frac{\chi^2_P}{M} = \frac{1}{M}\sum_i \frac{(I_i - F_i)^2}{F_i},$$

$$\frac{G}{M} = \frac{2}{M}\sum_i \left[I_i \ln\frac{I_i}{F_i} - (I_i - F_i)\right], \qquad D_{KL} = \sum_i p_i \ln\frac{p_i}{\hat{p}_i}, \quad p_i = \frac{I_i}{\sum_j I_j},\ \hat{p}_i = \frac{F_i}{\sum_j F_j}.$$

- Pearson's χ^2_P and the Poisson deviance (G-test) G are the χ^2 and the likelihood-ratio statistic for Poisson noise, $\sigma_i^2 = F_i$. Both are $\approx M$ for a correct model **only if** I is in counts. For intensities in other units they scale with the unit ($I \to cI$ multiplies both by c), so they cannot replace $\chi^2_r = 1$ as a target.
- D_{KL} of the normalized curves is scale-free (SASfit's G-test is $2D_{KL}$) and needs no errors, but it has no target value: it can monitor or compare fits, not choose a regularization parameter.

- Without σ_i a stopping target therefore needs a noise model (above), or a parameter choice that does not use the absolute error level: GCV, the L-curve, or the Bayesian evidence with an estimated error scale.

All four statistics are shown in the GUI's result panel. The EM solvers also store SASfit's G-test in `diagnostics["g_test_final"]`.

4.2 Discretization of the integral

4.2.1 General scheme

Every discretization represents the unknown by a finite expansion

$$x(r) = \sum_{k=1}^{N} c_k \, \varphi_k(r) \tag{4.6}$$

and turns (4.1) into the linear system $\mathbf{A}\mathbf{c} \approx \mathbf{b}$ with

$$A_{ik} = \int_0^{r_{max}} K(q_i, r) \, \varphi_k(r) \, \mathrm{d}r \qquad \text{(collocation in } q\text{)}. \tag{4.7}$$

The data are matched exactly at the measured q_i (collocation); Galerkin projections are not used because the data exist only at discrete q. Three choices of φ_k are implemented: point values with a simple quadrature, point values with interpolation and a fine quadrature, and cubic B-splines. The table summarizes when each is used.

representation	unknowns c_k	used when
direct collocation	$x(r_k)$	uniform grid with step $\leq 1.1\,\pi/q_{max}$, linear interpolation, B-splines off
nodes + interpolation, fine quadrature	$x(r_k)$	non-uniform grid, cubic interpolation, or uniform step $> 1.1\,\pi/q_{max}$ (automatic)
cubic B-splines	spline coefficients	"Represent p(r) by B-splines" checked

4.2.2 Direct collocation on a point grid

`sasfit_inversion.kernels.build_size_distribution_kernel()` uses the grid points themselves as quadrature nodes,

$$A_{ij} = \Delta r_j \; r_j^{-\alpha} \, K(q_i, r_j), \qquad \Delta r_j = \tfrac{1}{2}(r_{j+1} - r_{j-1}), \tag{4.8}$$

(`numpy.gradient`; one-sided at the ends). This is the rectangle/ trapezoid rule and is accurate only if the kernel varies little between neighbouring points. For the j_0 kernel the fastest oscillation in r has period $2\pi/q_{max}$, so a grid step $\Delta r > \pi/q_{max}$ undersamples the kernel at high q and *no* coefficient vector can fit those data. On test.dat ($\pi/q_{max} = 8.2$ nm, $D_{max} = 782$ nm) even unregularized least squares gives $\chi^2_r = 460$–840 for $N \leq 45$ points.

4.2.3 Point values with interpolation and fine quadrature

To decouple the number of unknowns from the quadrature accuracy, `build_kernel_on_nodes()` keeps the node values $c_k = x(r_k)$ as unknowns, interpolates between the nodes with fixed cardinal functions φ_k ($\varphi_k(r_j) = \delta_{jk}$), and evaluates (4.7) on a fine uniform grid ρ_l:

$$\mathbf{A} = \mathbf{A}^{fine}\,\mathbf{\Phi}, \qquad A^{fine}_{il} = \Delta\rho \, \rho_l^{-\alpha} K(q_i, \rho_l), \qquad \Phi_{lk} = \varphi_k(\rho_l). \tag{4.9}$$

The size of the fine grid (`quadrature_points()`) is

$$n_q = \max\left(4(N-1)+1,\ \left\lceil \frac{4\, r_{max}\, q_{max}}{\pi} \right\rceil + 1\right), \tag{4.10}$$

i.e. at least four points per Shannon interval π/q_{max} and at least four per node interval. The node kernel reproduces a reference integral of a sphere PDDF to 2.7×10^{-4} (relative) on a 60-point quadratic grid.

Linear interpolation (hat functions)

$$\varphi_k(r) = \begin{cases} \dfrac{r - r_{k-1}}{r_k - r_{k-1}} & r_{k-1} \le r \le r_k \\ \dfrac{r_{k+1} - r}{r_{k+1} - r_k} & r_k \le r \le r_{k+1} \\ 0 & \text{otherwise} \end{cases}$$

$\varphi_k \ge 0$ and $\sum_k \varphi_k = 1$: non-negative node values give a non-negative $x(r)$ and a non-negative kernel stays non-negative. Hence linear interpolation is compatible with every solver.

Cubic spline interpolation (not-a-knot)

φ_k is the C^2 cubic interpolating spline through the unit vector $\mathbf{e}_k$ (`cubic_basis()`). On each interval $x(r)$ is a cubic; x, x' and x'' are continuous at the inner nodes. The two missing conditions are the *not-a-knot* conditions – x''' is also continuous at r_2 and r_{N-1}, i.e. the first two and the last two intervals are each one cubic. The alternative *natural* condition $x''(0) = x''(r_{max}) = 0$ is wrong for a PDDF, which behaves like r^2 near the origin (test.dat, uniform 45 points: χ^2_r 2.42 natural vs 0.65 not-a-knot). The interpolation is exact for cubic polynomials and $\sum_k \varphi_k = 1$.

The cardinal cubic splines oscillate: they have negative lobes (minimum about -3.2 on a 60-point log grid, where adjacent intervals differ strongly in length). Consequently non-negative node values do **not** guarantee $x(r) \ge 0$, and a non-negative kernel becomes slightly signed (3 % negative entries for the sphere kernel). The GUI therefore uses linear interpolation for the solvers that need $x \ge 0$ or a non-negative kernel (EM + smoothing, EM + MaxEnt, Hansen MaxEnt, ARLS ≥ 0).

Effect of the representation (test.dat, j_0 kernel, Bayesian evidence, χ^2_r):

r grid	direct collocation	linear, fine quadrature	cubic, fine quadrature
uniform, 95 points	0.43	0.47	0.44
uniform, 45 points	–	13.9	0.65
uniform, 30 points	580	282	1.18
quadratic, 60 points	20.4	0.53	–
quadratic, 30 points	516	1.41	0.79
log, 60 points	96.2	0.65	0.60
log, 30 points	–	2.29	0.81

4.2.4 Spacing of the grid points

`make_r_grid()` places N points on $[0, r_{max}]$ with $u_k = (k-1)/(N-1)$:

$$\text{uniform: } r_k = r_{max}\, u_k, \qquad \text{quadratic: } r_k = r_{max}\, u_k^2,$$

$$\text{log: } r_1 = 0, \quad r_k = r_{max}\left(\frac{1}{2N}\right)^{\frac{N-k}{N-2}}, \; k = 2 \dots N,$$

(a geometric series from $r_{max}/(2N)$ to r_{max}). Dense spacing at small r suits a PDDF that rises steeply and has a long smooth tail; size distributions with structure at large r prefer the uniform grid.

All smoothness penalties of the next chapter are built from differences of *neighbouring unknowns*. On a non-uniform grid they therefore measure smoothness in the grid index – on a log grid, smoothness in $\ln r$, which gives the solution more freedom at small r.

4.2.5 Cubic B-spline basis

Following Glatter (1977) the unknown can be represented by $N_c \ll N_s$ cubic B-splines (`sasfit_inversion.spline_basis`):

$$x(r) = \sum_{k=1}^{N_c} c_k \, B_k(r/r_{max}). \tag{4.11}$$

Knot vector

Clamped, on $t = r/r_{max} \in [0, 1]$, degree $d = 3$:

$$\boldsymbol{\tau} = (\underbrace{0, 0, 0, 0}_{d+1}, \; \tau_1, \dots, \tau_K, \; \underbrace{1, 1, 1, 1}_{d+1}), \qquad K = N_c + 2 - d - 1.$$

The interior knots follow the same three rules as the grid points (`interior_knots()`): uniform $\tau_j = j/(K+1)$, quadratic $\tau_j = (j/(K+1))^2$, log geometric from $1/(2N_c)$ to 1.

Cox–de Boor recursion

$$B_{j,0}(t) = \begin{cases} 1 & \tau_j \le t < \tau_{j+1} \\ 0 & \text{else} \end{cases}, \qquad B_{j,d}(t) = \frac{t - \tau_j}{\tau_{j+d} - \tau_j} B_{j,d-1}(t) + \frac{\tau_{j+d+1} - t}{\tau_{j+d+1} - \tau_{j+1}} B_{j+1,d-1}(t).$$

The clamped basis has $N_c + 2$ functions; the first and the last (the only ones that are non-zero at $t = 0$ and $t = 1$) are dropped, which builds $x(0) = x(r_{max}) = 0$ into the basis instead of imposing it by a penalty.

Properties

$B_k \ge 0$, local support on four knot intervals, C^2 continuity. By the convex-hull property, $\mathbf{c} \ge 0$ implies $x(r) \ge 0$; a non-negative kernel stays non-negative. Every solver – including the EM and MaxEnt solvers – can therefore be used unchanged on the coefficients.

Kernel and results

With $\mathbf{\Phi}_{lk} = B_k(\rho_l/r_{max})$ on the fine grid (4.10) and $\mathbf{B}_{jk} = B_k(r_j/r_{max})$ on the display grid,

$$\mathbf{A}_{spline} = \mathbf{A}^{fine}\mathbf{\Phi} \in \mathbb{R}^{M \times N_c}, \qquad \mathbf{x} = \mathbf{B}\,\mathbf{c}, \qquad \mathrm{Cov}(\mathbf{x}) = \mathbf{B}\,\mathrm{Cov}(\mathbf{c})\,\mathbf{B}^T.$$

`sasfit_inversion.solver_registry.run_solver()` with `spline=SplineOptions(n_coeffs, spacing, A_quad)` runs any registered solver on $\mathbf{A}_{spline}$ and maps the result back. The solvers' smoothness operators then act on neighbouring coefficients.

Why it helps

A cubic B-spline basis with knot spacing $\Delta\tau\, r_{max}$ cannot represent oscillations shorter than about $2\Delta\tau\, r_{max}$. With $N_c = 20$ the basis cannot follow individual high-q points of

log-spaced data, while a 95-point grid can. On test.dat (j_0, 20 coefficients, log knots) all Bayesian-evidence and Tikhonov variants reach $\chi^2_r \approx 1$ with a high-q χ^2 of about 1.05, against 0.02–0.1 on the 95-point grid, and the Bayesian normalized evidence increases from $\ln Z = -153.0$ to -118.8. With the sphere kernel, the spurious spike at $r \approx 3$ nm of the volume-weighted point-grid solution disappears.

The number of coefficients is the regularization knob of this representation (typically 15–30); knot spacing matters as well (uniform knots needed about 25 coefficients on test.dat, log/quadratic knots 20).

4.2.6 Alternatives in the literature

- **Sine series** (Moore 1980): $p(r) = \sum_{n=1}^{N} a_n \sin(n\pi r/D_{max})$. The basis functions are the Shannon-optimal functions for a PDDF, vanish at both ends, and the integrals (4.7) with j_0 have closed forms. It is linear and signed (no positivity). Not implemented.
- **Shape-preserving cubic interpolation** (PCHIP, Fritsch & Carlson 1980) does not overshoot, but its slopes depend non-linearly on the node values, so it cannot be written as a fixed matrix $\mathbf{\Phi}$; not usable in the linear framework.
- **Higher-order quadrature** (Simpson, Gauss) of the collocation integrals improves the accuracy of (4.8) but not the undersampling problem, which is solved by the fine quadrature above.

4.3 Regularization and choice of the regularization strength

4.3.1 Weighted least squares

All linear methods work on the statistically weighted system

$$\tilde{\mathbf{A}} = \mathbf{W}^{1/2}\mathbf{A}, \qquad \tilde{\mathbf{b}} = \mathbf{W}^{1/2}\mathbf{b}, \qquad \|\tilde{\mathbf{A}}\mathbf{x} - \tilde{\mathbf{b}}\|^2 = \chi^2,$$

(`solver_registry._weighted`). Without the weights the few high-intensity low-q points dominate: on test.dat σ spans four decades (0.01 to 90).

Let $\tilde{\mathbf{A}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ with singular values $s_1 \geq s_2 \geq \ldots$. The naive solution $\mathbf{x} = \sum_i (\mathbf{u}_i^T\tilde{\mathbf{b}}/s_i)\,\mathbf{v}_i$ divides the noise component of $\mathbf{u}_i^T\tilde{\mathbf{b}}$ by singular values that decay to 10^{-15} (condition number $\sim 10^{16}$ for the j_0 kernel on test.dat). Regularization replaces $1/s_i$ by filtered values.

4.3.2 Tikhonov regularization

General form

$$\mathbf{x}_\lambda = \arg\min_{\mathbf{x}} \; \chi^2(\mathbf{x}) + \lambda^2\|\mathbf{L}\mathbf{x}\|^2 = \left(\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \lambda^2\mathbf{L}^T\mathbf{L}\right)^{-1}\tilde{\mathbf{A}}^T\tilde{\mathbf{b}}. \tag{4.12}$$

Standard form (`tikhonov_standard_gcv`)
$\mathbf{L} = \mathbf{I}$; in the SVD basis the filter factors are

$$f_i = \frac{s_i^2}{s_i^2 + \lambda^2}, \qquad \mathbf{x}_\lambda = \sum_i f_i \frac{\mathbf{u}_i^T\tilde{\mathbf{b}}}{s_i}\,\mathbf{v}_i.$$

Truncated SVD (`tsvd_gcv`)
$f_i = 1$ for $i \leq k$, 0 otherwise.

General form (`general_tikhonov_gcv`)

$\mathbf{L}$ is the second-difference operator below. A rank-deficient $\mathbf{L} \in \mathbb{R}^{p\times N}$ is handled by Hansen's QR-based transformation to standard form (`std_form`, method 1 of Regularization Tools): with $\mathbf{L}^T = \mathbf{K}\mathbf{R}$ ($\mathbf{K} = [\mathbf{K}_p, \mathbf{K}_0]$, $\mathbf{K}_0$ spanning the null space of $\mathbf{L}$) and $\tilde{\mathbf{A}}\mathbf{K}_0 = \mathbf{H}\mathbf{T}$,

$$\mathbf{L}_p^\dagger = \mathbf{K}_p\mathbf{R}^{-T}, \quad \bar{\mathbf{A}} = \mathbf{H}_2^T\tilde{\mathbf{A}}\mathbf{L}_p^\dagger, \quad \bar{\mathbf{b}} = \mathbf{H}_2^T\tilde{\mathbf{b}},$$

the standard-form problem in $\bar{\mathbf{x}}$ is solved with filter factors and mapped back by $\mathbf{x} = \mathbf{L}_p^\dagger\bar{\mathbf{x}} + \mathbf{K}_0\mathbf{T}^{-1}\mathbf{H}_1^T(\tilde{\mathbf{b}} - \tilde{\mathbf{A}}\mathbf{L}_p^\dagger\bar{\mathbf{x}})$. The result agrees with a direct solve of the normal equations to 10^{-11}.

4.3.3 Smoothness penalties

Let $p_1 \dots p_N$ be the unknowns (grid values or spline coefficients).

Second difference, free ends (`second_derivative`)

$$S_2(\mathbf{p}) = \sum_{j=2}^{N-1}(p_{j-1} - 2p_j + p_{j+1})^2 = \|\mathbf{L}_2\mathbf{p}\|^2, \qquad \mathbf{L}_2 \in \mathbb{R}^{(N-2)\times N}.$$

The null space of $\mathbf{L}_2$ contains constants and linear ramps: these components are not penalized at all. Together with the ill-conditioned j_0 kernel this gave χ^2_r in the hundreds on real data with GCV or the L-curve.

Hansen (2000), matrix as printed (`hansen_printed`)

The text after Hansen's eq. 19 gives the tridiagonal matrix $a_{ii} = 1$, $a_{i,i\pm1} = -\frac{1}{2}$, which is the Hessian of

$$S_{pr}(\mathbf{p}) = \tfrac{1}{2}\sum_{j=1}^{N-1}(p_j - p_{j+1})^2 + \tfrac{1}{2}p_1^2 + \tfrac{1}{2}p_N^2,$$

a *first-difference* penalty plus the assumption that p vanishes just outside $[0, D_{max}]$. It has full rank with $\det = (N+1)/2^N$ (checked at construction).

Hansen (2000) eq. 19 (`hansen_eq19`, default)

$$S_{19}(\mathbf{p}) = \sum_{j=2}^{N-1}\left[p_j - \tfrac{1}{2}(p_{j-1} + p_{j+1})\right]^2 + \tfrac{1}{2}p_1^2 + \tfrac{1}{2}p_N^2, \tag{4.13}$$

a *curvature* penalty with the same boundary terms; its Hessian $\mathbf{D}^T\mathbf{D} + \frac{1}{2}(\mathbf{e}_1\mathbf{e}_1^T + \mathbf{e}_N\mathbf{e}_N^T)$ ($\mathbf{D}$ with rows $(-\frac{1}{2}, 1, -\frac{1}{2})$) is five-diagonal and positive definite. The printed matrix and eq. 19 are therefore *not* the same penalty; both are offered. The normalized evidence (below) prefers eq. 19 on test.dat ($\ln Z = -153.0$ vs -236.2 on the 95-point grid, -118.8 vs -134.1 with 20 B-splines), which made it the default.

In all three cases $\mathbf{L}$ is a factor with $\mathbf{L}^T\mathbf{L}$ = Hessian of S (Cholesky factor for the Hansen penalties).

4.3.4 Bayesian evidence (`bayesian_evidence`)

Following Hansen (2000) and Vestergaard & Hansen (2006) the regularization is read as a Gaussian prior and the likelihood as Gaussian noise:

$$P(\mathbf{p} \mid \lambda) \propto \exp(-\lambda S(\mathbf{p}))\,, \qquad P(\mathbf{d} \mid \mathbf{p}) \propto \exp\!\left(-\tfrac{1}{2}\chi^2(\mathbf{p})\right), \qquad S(\mathbf{p}) = \mathbf{p}^T\mathbf{L}^T\mathbf{L}\,\mathbf{p}.$$

Posterior maximum (MAP)

$$\mathbf{p}_\lambda = \arg\min\ \lambda S + \tfrac{1}{2}\chi^2 = \left(2\lambda \mathbf{L}^T\mathbf{L} + \mathbf{B}\right)^{-1} \mathbf{A}^T\mathbf{W}\mathbf{b}, \qquad \mathbf{B} = \mathbf{A}^T\mathbf{W}\mathbf{A}. \tag{4.14}$$

($\mathbf{B}$ is the Hessian of $\chi^2/2$, $\mathbf{A}_S = 2\mathbf{L}^T\mathbf{L}$ that of S.)

Evidence

Integrating the posterior over $\mathbf{p}$ (exact for this linear-Gaussian model) gives the probability of the data for a given λ:

$$\ln P(\mathbf{d} \mid \lambda) = \tfrac{1}{2}\ln\det(\lambda\mathbf{A}_S) - \tfrac{1}{2}\ln\det(\lambda\mathbf{A}_S + \mathbf{B}) - \lambda S(\mathbf{p}_\lambda) - \tfrac{1}{2}\chi^2(\mathbf{p}_\lambda) + \text{const.} \tag{4.15}$$

For the λ search the first term is replaced by $\frac{N-1}{2}\ln\lambda$ (Vestergaard & Hansen eq. 6; the remaining λ-independent part of $\det\mathbf{A}_S$ does not move the maximum). λ is the maximizer on a log grid of 60 values spanning $10^{\pm 6}\,\text{median}(s_i)^2$ (s_i singular values of $\mathbf{A}$).

Normalized evidence (penalty comparison)

To compare *different* penalties, the full prior normalization must be kept:

$$\ln Z = \tfrac{1}{2}\ln\det(2\lambda\mathbf{L}^T\mathbf{L}) - \tfrac{1}{2}\ln\det(2\lambda\mathbf{L}^T\mathbf{L} + \mathbf{B}) - \lambda S - \tfrac{1}{2}\chi^2.$$

It is reported in the result line. For the second difference $\mathbf{L}^T\mathbf{L}$ is singular (improper prior) and $\ln Z$ is not defined.

Unknown error scale (`noise_scale="estimate"`)

If the errors are known only up to a common factor, $\sigma_i \to s\,\sigma_i$, the likelihood normalization adds $-M\ln s$ to (4.15) and s is a second hyperparameter. `sasfit_inversion.solver_registry.estimate_noise_scale()` maximizes

$$E(s) = \max_\lambda \ln P(\mathbf{d} \mid \lambda, s\boldsymbol{\sigma}) - M\ln s$$

over $\ln s \in [\ln 0.01, \ln 100]$ (bounded scalar search, the inner maximum on an 80-point λ grid over 16 decades) and then solves with $s\boldsymbol{\sigma}$. This is the Bayesian analogue of estimating the noise level from the residuals corrected for the number of fitted parameters (MacKay: at the optimum $\chi^2(s\boldsymbol{\sigma}) = M - \gamma$, γ the effective number of parameters determined by the data). test.dat: with $3\Delta I$ or $\Delta I/3$ the estimate is $s = 0.323$ and 2.91 (product 0.97), and $p(r)$ is identical to the one with the file errors (correlation 1.0000); with errors from the local cubic fits $s = 0.91$. χ^2_r is reported for the unscaled σ.

Number of good parameters

$$N_g = \sum_j \frac{\mu_j}{\lambda + \mu_j},$$

with μ_j the eigenvalues of $\mathbf{B}$ (the $\mathbf{A}_S = \mathbf{I}$ form of Vestergaard & Hansen's eq. 8), an estimate of the number of parameters determined by the data; compare with N_s of (4.5).

Posterior covariance

The Laplace approximation, exact here, gives $\text{Cov}(\mathbf{p}) = (\lambda\mathbf{A}_S + \mathbf{B})^{-1}$, shown as the analytic error band (*Uncertainty of the recovered distribution*).

4.3.5 Discrepancy principle

Choose the regularization parameter θ (λ, λ^2, the EM smoothing weight h or the entropy weight) such that

$$\chi^2_r(\theta) = 1,$$

i.e. the residuals are as large as the stated errors. Two searches are implemented (`sasfit_inversion.solvers.lambda_search`):

- `discrepancy_principle_search` starts over-smoothed and walks θ down geometrically until $\chi_r^2 - 1$ changes sign, then refines by Brent's method (EM + smoothing).
- `discrepancy_principle_search_bidirectional` decides the direction from the sign of $\chi_r^2 - 1$ at the start value, walks geometrically until the root is bracketed and refines by Brent. If the target is not reachable (plateau of χ_r^2 for four steps, or the search range is exhausted) the closest achievable solution is returned with `reachable=False` and a warning in the result line, stating which side failed: a too-small basis (cannot fit) or overestimated errors (always fits better than the target).

For the Bayesian-evidence solver "Force $\chi_r^2 = 1$" starts at the evidence-maximizing λ_{ev} and allows λ to decrease to at most $10^{-3}\lambda_{ev}$. Reaching $\chi_r^2 = 1$ only by almost removing the regularization means that the target is wrong for these data (underestimated σ, or bootstrap replicates with inflated noise) and would give an oscillating, unregularized $p(r)$.

When the discrepancy principle fails for EM + smoothing on a point grid, the L-curve corner is used instead; in B-spline mode $h = 0$ is used (the basis regularizes already, and the L-curve picks far too much extra smoothing there).

4.3.6 L-curve

For a decreasing sequence $\theta_k = \theta_0/1.2^k$ the points $P_k = (\ln \rho_k, \ln \eta_k)$ with residual measure $\rho = \chi_r^2$ and roughness $\eta = \sum_j (x_{j+1} - x_j)^2$ are computed, and the corner is the point of maximum Menger curvature of consecutive triples,

$$\kappa(P_{k-1}, P_k, P_{k+1}) = \frac{4\,\mathcal{A}(P_{k-1}, P_k, P_{k+1})}{|P_{k-1}P_k|\,|P_kP_{k+1}|\,|P_{k-1}P_{k+1}|},$$

$\mathcal{A}$ the triangle area. This mirrors the `Lcorner` branch of SASfit's `DR_EM_smoothing`.

4.3.7 Generalized cross-validation

GCV chooses λ (or the TSVD rank k) by minimizing

$$G(\lambda) = \frac{\|\tilde{\mathbf{A}}\mathbf{x}_\lambda - \tilde{\mathbf{b}}\|^2}{(M - \sum_i f_i)^2},$$

the leave-one-out prediction error in rotation-invariant form (for the general form evaluated in the standard-form variables, for TSVD with $\sum_i f_i = k$). Grid: 100 values from $10^{-3}s_{min}$ to $10\,s_{max}$; TSVD $k \le 40$.

4.3.8 ARLS (`arlsnn`)

Jones' automatically regularized least squares uses the discrete Picard condition: the Tikhonov coefficients $g_i(\lambda) = \beta_i s_i/(s_i^2 + \lambda^2)$, $\boldsymbol{\beta} = \mathbf{U}^T\tilde{\mathbf{b}}$, should decay. The smallest λ for which a line and a parabola fitted to $\log_{10}|g_i|$ both have non-positive slope is taken, then λ is increased as long as the residual stays below twice its value. Non-negativity: columns with negative coefficients are removed repeatedly. No parameter has to be chosen; it serves as a quick sanity check.

4.4 EM, maximum entropy and signed kernels

4.4.1 Expectation–maximization (Richardson–Lucy)

For a non-negative kernel $A_{ij} \geq 0$ and non-negative unknowns, the EM algorithm maximizes the Poisson log-likelihood

$$\ell(\mathbf{x}) = \sum_i \left[b_i \ln(\mathbf{A}\mathbf{x})_i - (\mathbf{A}\mathbf{x})_i \right],$$

equivalently it minimizes the Kullback–Leibler divergence $\mathrm{KL}(\mathbf{b} \,\|\, \mathbf{A}\mathbf{x})$. The stationarity condition $\partial \ell / \partial x_j = 0$ for $x_j > 0$,

$$\sum_i A_{ij} \frac{b_i}{(\mathbf{A}\mathbf{x})_i} = \sum_i A_{ij},$$

is turned into the multiplicative fixed-point iteration (Kohlbrecher & Bressler 2022, eq. 36–37)

$$x_j^{(k+1)} = x_j^{(k)} \, \frac{\displaystyle\sum_i A_{ij}\, b_i / (\mathbf{A}\mathbf{x}^{(k)})_i}{\displaystyle\sum_i A_{ij}}. \tag{4.16}$$

Each step keeps $\mathbf{x} \geq 0$ and does not decrease ℓ. Columns with $\sum_i A_{ij} = 0$ (e.g. $r = 0$ with $\alpha > 0$) are left unchanged. The iteration converges slowly and, run to convergence without regularization, ends in a noisy maximum likelihood solution; early stopping acts as an uncontrolled regularizer ("semi-convergence"). All EM solvers here therefore run to convergence and regularize explicitly.

Stopping criterion: $\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2 \leq$ `tol` (SASfit's `gNorm`), for the accelerated iteration the relative change of $\|\mathbf{x}\|$.

4.4.2 EM + smoothing (`em_discrepancy`)

After every EM step the solution is smoothed with the tridiagonal operator (eq. 41–46)

$$\mathbf{S}^{(h)} = \begin{pmatrix} 1-h & h & & & \\ h & 1-2h & h & & \\ & \ddots & \ddots & \ddots & \\ & & h & 1-2h & h \\ & & & h & 1-h \end{pmatrix}, \qquad \mathbf{x}^{(k+1)} = \mathbf{S}^{(h)}\, \mathrm{EM}(\mathbf{x}^{(k)}),$$

$0 < h \lesssim 0.3$; rows sum to one, so the smoothing conserves $\sum_j x_j$ and positivity. (The optional *double smoothing* also smooths $\ln \mathbf{x}$ before the step, $\mathbf{x} \leftarrow \exp(\mathbf{S}^{(h)} \ln \mathbf{x})$.) The fixed point balances the EM correction against the diffusion-like smoothing; h plays the role of λ. It is chosen by the discrepancy principle (start $h = 0.3$, floor 10^{-9}) or the L-curve corner.

4.4.3 Biggs–Andrews acceleration

All EM-type iterations are accelerated by vector extrapolation (Biggs & Andrews 1997; `sasfit_inversion.solvers.acceleration.biggs_andrews()`, ported from SASfit's fixed-point solver). With the fixed-point map $\mathcal{T}$ and the last updates $\mathbf{g}_n = \mathcal{T}(\mathbf{y}_n) - \mathbf{y}_n$:

$$\alpha_n = \frac{\mathbf{g}_{n-1} \cdot \mathbf{g}_{n-2}}{\mathbf{g}_{n-2} \cdot \mathbf{g}_{n-2}}, \qquad |\alpha_n| \leq \frac{n-1}{n+2},$$

$$\mathbf{y}_n = \mathbf{x}_n + \alpha_n(\mathbf{x}_n - \mathbf{x}_{n-1}) + \tfrac{1}{2}\alpha_n^2(\mathbf{x}_n - 2\mathbf{x}_{n-1} + \mathbf{x}_{n-2}), \qquad \mathbf{x}_{n+1} = \mathcal{T}(\mathbf{y}_n),$$

(second order, `kin_set_maa=2`; order 1 drops the last term, order 0 is plain Picard iteration). For non-negative problems $\mathbf{y}_n$ is clipped to ≥ 0 before $\mathcal{T}$ is applied. The fixed point is unchanged; on the JAC 2022 benchmark the iteration count drops from more than 200 000 to about 2 800.

4.4.4 EM + MaxEnt (`maxent_em`)

SASfit's DR_EM_ME (Kohlbrecher & Bressler 2022, eq. 47–55) adds an entropy correction to each EM step,

$$S(\mathbf{x}, \mathbf{m}) = \sum_j \left[-x_j \ln \frac{x_j}{m_j} + x_j - m_j \right], \qquad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta\mathbf{x}_{EM}^{(k)} + \Delta\mathbf{s}^{(k)}.$$

Constant prior
$m_j = \bar{b}/N$ fixed and

$$\Delta s_j = \lambda\, x_j \left[-\ln \frac{x_j}{m_j} + \frac{\sum_l x_l \ln(x_l/m_l)}{\sum_l x_l} \right],$$

the gradient of S in the EM metric (multiplied by x_j), projected such that $\sum_j \Delta s_j = 0$.

Adaptive prior
The prior is the current solution smoothed with a Gaussian point spread function (Horne 1985),

$$m_i = \sum_j \Pi_{ij} x_j, \qquad \Pi_{ij} = \frac{1}{c_i} \exp\left[-\frac{(i-j)^2}{2\sigma^2} \right], \quad \sum_j \Pi_{ij} = 1,$$

($\sigma^2 = 1$ grid step squared) and

$$\Delta s_j = \lambda\, x_j \left[\frac{\sum_l x_l \ln(x_l/m_l)}{\sum_l x_l} - \ln \frac{x_j}{m_j} - 1 + \sum_i \Pi_{ij} \frac{x_i}{m_i} \right].$$

Entropy relative to a smoothed version of itself acts as a smoothness prior.

λ is chosen from 24 values between 10^{-9} and 1 as the one with χ^2_r closest to 1, every trial run to convergence.

4.4.5 Maximum entropy, Hansen's functional (`hansen_maxent`)

Following Hansen & Elfving (`maxent.m` of Regularization Tools) but with statistical weights and a flat default model m:

$$F(\mathbf{x}) = \sum_i \frac{[(\mathbf{A}\mathbf{x})_i - b_i]^2}{\sigma_i^2} + \lambda^2 \sum_j x_j \ln \frac{x_j}{m}, \qquad x_j > 0. \tag{4.17}$$

$$\nabla F = 2\tilde{\mathbf{A}}^T(\tilde{\mathbf{A}}\mathbf{x} - \tilde{\mathbf{b}}) + \lambda^2 \left(1 + \ln \frac{\mathbf{x}}{m}\right), \qquad \nabla^2 F = 2\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \lambda^2 \operatorname{diag}(1/x_j).$$

- Default model: the constant that best fits the data, $m = \frac{\mathbf{a}^T\tilde{\mathbf{b}}}{\mathbf{a}^T\mathbf{a}}$, $a_i = \sum_j \tilde{A}_{ij}$.
- Minimization: bound-constrained L-BFGS-B ($x_j \geq 10^{-30} m$) in variables scaled by $s_j = \sqrt{(\nabla^2 F)_{jj}}$ at $\mathbf{x} = m$, followed by a Newton polish on the free components (fraction-to-boundary 0.99, Armijo backtracking). Without the polish L-BFGS-B stops in the flat valley of the ill-conditioned j_0 problem with χ^2_r off by several percent and non-monotonic in λ.

- λ^2 by the bidirectional discrepancy principle, warm- starting each trial from the previous solution.

The original non-linear CG transcription of `maxent.m` (kept as `hansen_maxent.solve`) caps each step where the first component reaches zero and has no active-set handling; on test.dat it stopped after 2–18 iterations at $F = 71\,049$ where the minimum of the same functional is $F = 165$.

4.4.6 EM for signed kernels (`em_general_signed_kernel`)

The j_0 kernel and signed $p(r)$ violate both assumptions of (4.16). Chae, Martin & Walker (2019, Sec. 6) reduce the general problem to a non-negative one in two steps.

Shift (signed solution, their eq. 10)
With a constant $t > \max|p|$,

$$\tilde{p} = p + t \geq 0, \qquad \tilde{b}_i = b_i + t\sum_j |A_{ij}|,$$

Kernel split (signed kernel, their eq. 12)
$\mathbf{A} = \mathbf{A}^+ - \mathbf{A}^-$ with $\mathbf{A}^\pm \geq 0$ (pointwise positive and negative parts), extended kernel and unknown

$$\mathbf{K} = \begin{bmatrix}\mathbf{A}^+ & \mathbf{A}^-\end{bmatrix} \in \mathbb{R}^{M\times 2N}_{\geq 0}, \qquad \mathbf{u} = \begin{pmatrix}\mathbf{p}+t \\ -\mathbf{p}+t\end{pmatrix} \geq 0, \qquad \mathbf{K}\mathbf{u} = \tilde{\mathbf{b}}.$$

EM (4.16) is applied to $(\mathbf{K}, \tilde{\mathbf{b}})$. The extended problem has more solutions than those of structure $[\mathbf{p}, -\mathbf{p}]$, so after each step the structure is restored from the first half, $\mathbf{p} \leftarrow \mathbf{u}_{1:N} - t$ (an antisymmetrized variant is available). $|p_j|$ is clipped to $0.999\,t$. Default $t = 10\max_j |p_j^{ARLS}|$ from a quick ARLS estimate; results are nearly independent of t once it exceeds $\max|p|$.

χ^2 row weighting (2026-10-10)
EM minimizes $\mathrm{KL}(\tilde{\mathbf{b}}\|\mathbf{K}\mathbf{u}) \approx \sum_i (\mathbf{K}\mathbf{u} - \tilde{\mathbf{b}})_i^2/(2\tilde{b}_i)$: the implicit weights are $1/\tilde{b}_i$. After the shift $\tilde{b}_i \approx t\sum_j |A_{ij}|$ is dominated by the shift; for j_0 the row sums do not decay with q, and the real signal at high q is only 10^{-5}–10^{-4} of $\tilde{b}_i$ – EM is blind to these data (test.dat: $\chi^2_r \geq 5200$ without smoothing, for any t). Scaling each row by

$$w_i = \frac{\tilde{b}_i}{\sigma_i^2}: \qquad \mathbf{K} \to \mathrm{diag}(w)\mathbf{K}, \quad \tilde{\mathbf{b}} \to \mathrm{diag}(w)\tilde{\mathbf{b}},$$

keeps the kernel non-negative and changes the misfit to

$$\sum_i \frac{w_i^2(\mathbf{K}\mathbf{u} - \tilde{\mathbf{b}})_i^2}{2w_i\tilde{b}_i} = \sum_i \frac{(\mathbf{K}\mathbf{u} - \tilde{\mathbf{b}})_i^2}{2\sigma_i^2} = \tfrac{1}{2}\chi^2(\mathbf{p}),$$

because $\mathbf{K}\mathbf{u} - \tilde{\mathbf{b}} = \mathbf{A}\mathbf{p} - \mathbf{b}$. The fixed point then fits the data within their errors.

Acceleration safeguard
With the first-half rebuild the iteration is no longer a monotone EM map, and Biggs–Andrews extrapolation can drift (13 spline coefficients: χ^2_r 1.68 to 260). If the accelerated run ends unconverged with $\chi^2_r > 1.05\times$ the best iterate, plain Picard iteration continues from the best iterate (`acceleration_fallback` in the diagnostics).

Regularization options

smoothing – $\mathbf{p} \leftarrow \mathbf{S}^{(h)}\mathbf{p}$ after each step, h by the discrepancy principle (floor 10^{-7}).

MaxEnt – the entropy of a signed p is defined through the two-channel representation $p = h_+ - h_-$, $h_\pm \geq 0$, with default model m for both channels (Skilling & Gull); eliminating the channels gives

$$S_\pm(\mathbf{p}) = \sum_j \left[\psi_j - 2m - p_j \operatorname{asinh}\frac{p_j}{2m}\right], \qquad \psi_j = \sqrt{p_j^2 + 4m^2}, \qquad \frac{\partial S_\pm}{\partial p_j} = -\operatorname{asinh}\frac{p_j}{2m}.$$

In analogy to SASfit's $\Delta s = \lambda\, x\, \partial S/\partial x$ the metric ψ_j is used, and each EM step is followed by

$$p_j \leftarrow p_j + \lambda\,\psi_j\left[\tau_j - \operatorname{asinh}\frac{p_j}{2m}\right], \qquad \tau_j = \begin{cases} 0 & \text{constant prior} \\ \operatorname{asinh}\dfrac{(\mathbf{\Pi p})_j}{2m} & \text{adaptive prior} \end{cases}$$

For $|p| \ll m$ this is a ridge pull $-\lambda p$ (towards 0 or towards $\mathbf{\Pi p}$), for $|p| \gg m$ the usual MaxEnt behaviour $-\lambda|p|\ln(|p|/m)$. Default $m = 0.1\max|p^{ARLS}|$; λ by the bidirectional discrepancy principle (start 10^{-5}, range 10^{-12}–1). A first attempt applying SASfit's correction to the shifted $\mathbf{u}$ with prior t failed: the entropy then degenerates to a ridge with scale t, and the adaptive variant diverged.

Results (test.dat, j_0, correlation with the Bayesian-evidence $p(r)$):

setup	regularization	χ^2_r	correlation
95-point grid	smoothing	1.01	0.994
95-point grid	MaxEnt, adaptive	1.00	0.994
95-point grid	MaxEnt, constant	not converged	≈ 0.93
20 log B-splines	smoothing	1.00	0.9998
20 log B-splines	MaxEnt, adaptive	1.00	0.9998
20 log B-splines	MaxEnt, constant	0.99	0.9996
13 log B-splines (n = 15)	smoothing	1.87	0.995

The constant prior pulls towards zero, which does not damp the oscillating components that the data do not determine (slow null-space modes), and converges slowly on point grids. With the sphere kernel the signed EM fits ($\chi^2_r = 1$) but gives a clearly negative $N(R)$ (minimum $\approx -0.6\times$ maximum); for size distributions the positive EM solvers are the right choice.

4.5 Uncertainty of the recovered distribution

4.5.1 Analytic (Laplace) error band

For the Bayesian-evidence solver the posterior is Gaussian, with covariance equal to the inverse Hessian of $\lambda S + \chi^2/2$ at the MAP point:

$$\mathrm{Cov}(\mathbf{p}) = \left(2\lambda\mathbf{L}^T\mathbf{L} + \mathbf{A}^T\mathbf{W}\mathbf{A}\right)^{-1}, \qquad \sigma_{p_j} = \sqrt{\mathrm{Cov}_{jj}}.$$

It contains the noise propagation *and* the prior's uncertainty for a fixed λ, but not the uncertainty of λ itself. In B-spline mode the coefficient covariance is mapped to the display grid with $\mathbf{B}\,\mathrm{Cov}(\mathbf{c})\,\mathbf{B}^T$. The band is shown automatically ("MAP solution" with $\pm 1\sigma$).

4.5.2 Parametric bootstrap

`sasfit_inversion.uncertainty.bootstrap_uncertainty()` works for every solver (including the non-linear EM and MaxEnt solvers and the automatic choice of the regularization parameter, which is repeated for every replicate):

$$\mathbf{b}^{(k)} = \hat{\mathbf{b}} + \boldsymbol{\varepsilon}^{(k)}, \qquad \varepsilon_i^{(k)} \sim \mathcal{N}(0, \sigma_i^2), \qquad k = 1 \dots N_{boot},$$

$$\bar{\mathbf{x}} = \frac{1}{N_{boot}} \sum_k \mathbf{x}^{(k)}, \qquad \sigma_{x_j}^2 = \frac{1}{N_{boot}} \sum_k \left(x_j^{(k)} - \bar{x}_j \right)^2 .$$

Replicates whose solve fails are skipped and counted. The relative error of σ_{x_j} is about $1/\sqrt{2N_{boot}}$ (10 % for $N_{boot} = 50$).

Centre of the replicates

The replicates are generated around the **fitted** curve $\hat{\mathbf{b}} = \mathbf{A}\mathbf{x}$ of the single solve, not around the measured $\mathbf{b}$. The measured data already contain one realization of noise; $\mathbf{b} + \boldsymbol{\varepsilon}$ has variance $2\sigma_i^2$ and every replicate a true $\chi_r^2 \approx 2$. A solver forced to $\chi_r^2 = 1$ must then remove most of its regularization on such replicates: on test.dat centring on $\mathbf{b}$ drove λ down by up to 10^8 (std of p up to 5.9 for a peak of 1), centring on the fit gave std ≤ 0.014. The floor $\lambda \geq 10^{-3}\lambda_{ev}$ of the forced-χ^2 search is a second safeguard.

The GUI plots the bootstrap mean with a $\pm 1\sigma$ band and the single solve as a dashed line; for non-linear solvers the two need not coincide.

4.6 Development history and findings

This chapter keeps the reasoning behind the present methods – including what was tried and discarded – for anyone extending the package. All numbers refer to `tests/data/test.dat` (bilog-normal spheres with modes near 60 and 180 nm, q in nm^{-1}, 91 log-spaced points, background ≈ 0.1) unless stated otherwise.

4.6.1 The null space of the second-difference penalty

With the j_0 kernel and the plain second-difference penalty, GCV, the L-curve and the evidence all gave χ_r^2 in the hundreds (EM + smoothing 586, Tikhonov + GCV 34 600). Constants and linear ramps are not penalized, and with a kernel condition number of $\sim 10^{16}$ these free directions are enough to make every criterion choose an oscillating solution. Hansen's (2000) boundary terms remove the null space (condition number of $\mathbf{L}^T\mathbf{L} \sim 9 \times 10^3$) and gave a clean evidence maximum.

4.6.2 Regional overfitting at high q

A good overall χ_r^2 hid an unbalanced fit: with the Hansen penalty on the 95-point grid the high-q third of the data had $\chi^2 \approx 0.02$ (the fit followed the noise), the low-q third about 0.8. Scans that each held everything else fixed ruled out the obvious causes:

- λ: no value gave all three regions near 1 at once.
- D_{max}: 500 nm was too small (low-q χ^2 3.35 to 0.8 up to $D_{max} \approx 800$), which led to the suggestion $D_{max} = \pi/q_{min}$; the high-q value stayed at 0.02 for every D_{max}.
- number of grid points above ≈ 75 and r_{min}: no effect.

Two attempted fixes, now retired:

- **Two-segment** λ (independent λ for low and high r, chosen by a two-hyperparameter evidence search): within 0.3–0.6 % of the single-λ result, never improved the high-q fit. The j_0 kernel couples every data point to the whole r range, so the noise-following component is not localized in r.
- **Hann-tapered boundary** ($p = w(r)\, p_{free}$ with a raised-cosine roll-off over the last quarter of $[0, D_{max}]$): moved the high-q χ^2 from 0.023 to 0.21, but needed a ridge stabilizer, and was superseded by the B-spline representation.

The explanation is Shannon undersampling of log-spaced data (*Forward model, data and information content*): at high q there are only 0.45 data points per Shannon channel, so a model with one free parameter per channel can follow every point. Reducing the number of free parameters with a B-spline basis (*Discretization of the integral*) brings the high-q χ^2 to about 1 for every solver and increases the normalized evidence ($\ln Z$ from -153.0 to -118.8). The retired solvers remain callable by their old keys (`RETIRED_SOLVERS`) but are no longer offered.

4.6.3 Bootstrap with forced $\chi^2_r = 1$

The bootstrap exploded (std of p up to 400) when combined with forced $\chi^2_r = 1$, because the replicates were centred on the noisy data. Fixed by centring on the fit and by the λ floor (*Uncertainty of the recovered distribution*).

4.6.4 Hansen MaxEnt

The CG transcription of `maxent.m` jammed at the positivity bound and the registry wrapper ignored σ_i; replaced by weighted L-BFGS-B with a Newton polish and the discrepancy principle (*EM, maximum entropy and signed kernels*).

4.6.5 Signed EM

Three problems were found and fixed on 2026-10-10: the shifted KL misfit was blind to the high-q data (fixed by χ^2 row weighting), Biggs–Andrews extrapolation could drift away from the fixed point (safeguard), and too coarse quadrature with few grid points ((4.10)). MaxEnt regularization with a positive/negative entropy was added as an alternative to smoothing.

4.6.6 Few grid points

A uniform point grid with step $> \pi/q_{max}$ undersamples the j_0 kernel (χ^2_r 460–840 for $N \leq 45$); this affects every solver. The fine quadrature of the node and spline representations removes the limitation; with cubic interpolation 30 uniform points suffice (χ^2_r 1.18).

4.6.7 Reduced solver list

Solvers that differed in a single setting were merged and given options (e.g. EM + smoothing with discrepancy principle or L-curve, Bayesian evidence with three penalties); the B-spline solvers were replaced by the generic B-spline option. Old keys are mapped by `SOLVER_ALIASES`.

4.6.8 Non-negativity is not a default

$p(r) \geq 0$ holds only for homogeneous contrast. For core–shell, multilayer or contrast-variation systems $p(r)$ has negative parts that carry structural information. Non-negativity is therefore only used where it is part of the method (EM, MaxEnt, ARLS ≥ 0) and never imposed on the signed solvers.

4.6.9 Open questions

- Automatic choice of the number of B-spline coefficients and of the knot spacing (e.g. by the normalized evidence).
- Penalties scaled by the node spacing (smoothness in r rather than in the index) for non-uniform point grids.
- The default level m of the signed MaxEnt is a heuristic.
- Moore's sine-series basis for the PDDF.
- Whether the quoted σ_i of test.dat are well calibrated at high q.

4.7 Notation

$q_i,\ i = 1 \dots M$	scattering vector magnitudes of the M data points
$b_i,\ \sigma_i$	background-subtracted intensity $I(q_i) - I_{bg}$ and its standard uncertainty $\Delta I(q_i)$
r, D_{max}, r_{max}	real-space variable (pair distance or radius), maximum particle dimension, upper end of the r grid
$p(r)$, $x(r)$	unknown function: pair-distance distribution (PDDF) or weighted size distribution $x(R) = N(R)\,R^{\alpha}$
$K(q, r)$	kernel of the Fredholm integral
$\mathbf{A} \in \mathbb{R}^{M \times N}$, $\mathbf{x}$	discretized kernel and vector of unknowns (point values or basis coefficients)
$\mathbf{W} = \mathrm{diag}(1/\sigma_i^2)$	statistical weights
χ^2, χ^2_r	$\chi^2 = \sum_i \left[(\mathbf{Ax})_i - b_i\right]^2 / \sigma_i^2$; in this package $\chi^2_r = \chi^2 / M$ (no subtraction of the number of parameters, as in SASfit's C code)
$\mathbf{L}$, λ	regularization (penalty) operator and its weight

4.8 Main references

The key papers are Glatter (1977) for the B-spline IFT, Moore (1980) for the information content, Hansen (2000) and Vestergaard & Hansen (2006) for the Bayesian IFT, Kohlbrecher & Breßler (2022) for SASfit's EM solvers, Chae, Martin & Walker (2019) for EM with signed kernels, Silverman *et al.* (1990) for EM with smoothing, and Hansen's Regularization Tools. The complete, topic-ordered list with notes on where each work is used is in *References*.

CHAPTER

FIVE

REFERENCES

Literature relevant for the methods implemented in *sasfit-inversion*, grouped by topic. Each entry notes where it is used in the package or in this documentation. DOIs are given where they were checked.

5.1 Small-angle scattering: indirect Fourier transformation and size distributions

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CHAPTER

SIX

INDICES AND TABLES

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